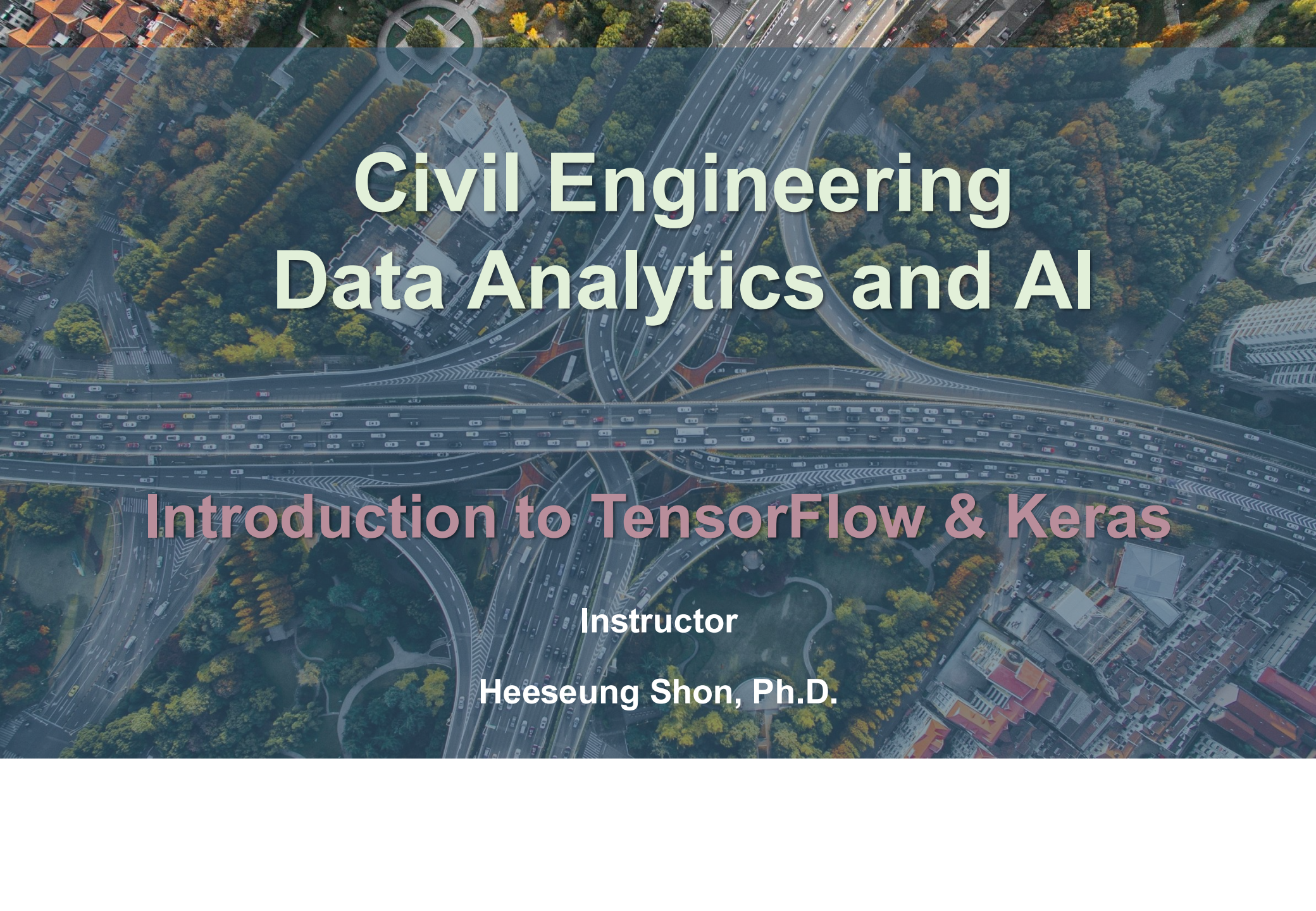




Civil Engineering Data Analytics and AI

Introduction to TensorFlow & Keras

Instructor

Heeseung Shon, Ph.D.

Deep Learning Frameworks

❑ Evolution of Deep Learning Frameworks

- ❖ Deep learning frameworks simplify the development, training, evaluation, and deployment of neural network models.

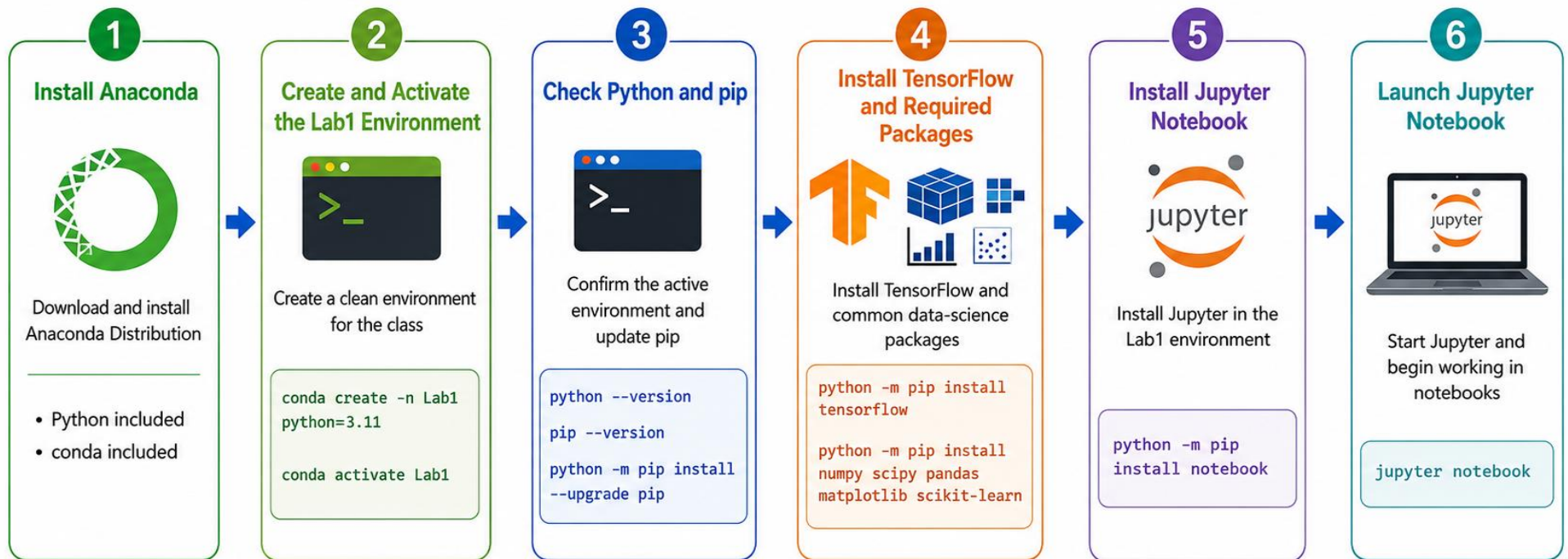


Deep Learning Frameworks

Framework / Ecosystem	Main language	Strength	Learning difficulty	Typical use
TensorFlow / Keras 3	Python	Deployment ecosystem	Easy-Moderate	Education and production
PyTorch	Python, C++	Flexibility and research	Easy-Moderate	Research and custom models
JAX / Flax	Python	High-performance numerical computing	Difficult	Advanced research
Deeplearning4j	Java, Scala	JVM and enterprise integration	Moderate-Difficult	Java-based systems

How to Use TensorFlow & Keras

❖ Using Anaconda + Jupyter Notebook (Python)



TensorFlow & Keras Setup

❑ Step 1. Install Anaconda

- ❖ Download Anaconda Distribution
- ❖ <https://www.anaconda.com/download>

Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Package and environment management with conda, Anaconda Navigator desktop app, and 600+ packages. Everything you need for data science.

GRAPHICAL INSTALLER

[↓ 64-Bit \(Apple silicon\) Graphical Installer](#)

📘 Graphical install instructions

COMMAND LINE

📄 CLI install instructions

Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

GRAPHICAL INSTALLER

[↓ 64-Bit \(Apple silicon\) Graphical Installer](#)

📘 Graphical install instructions

COMMAND LINE

📄 CLI install instructions

TensorFlow & Keras Setup

❑ Step 2. Python and TensorFlow Requirements

- ❖ If you use Anaconda, Python is already included.
- ❖ Recommended classroom setup: Python 3.10 or 3.11
- ❖ Official TensorFlow pip support: Python 3.9-3.12
- ❖ Update pip before installing TensorFlow

TensorFlow & Keras Setup

❑ Step 3. Create a New Environment

The screenshot shows the Anaconda Navigator application window. The left sidebar contains navigation options: Home, Environments (highlighted with a red dashed box and a red '1'), Learning, and Community. Below these is a 'Launch Your Environment' button and links to Documentation and Anaconda Blog. At the bottom of the sidebar, there are icons for X, GitHub, YouTube, and LinkedIn, and a 'Create' button (highlighted with a red dashed box and a red '2'). The main panel displays a list of environments: base (root), BRL, LLM_research_AAM, and UAM_Research. A 'Create new environment' dialog box is open in the center, with fields for Name (Lab1), Location (/opt/anaconda3/envs/Lab1), and Packages (Python 3.11.15, R). The dialog box has 'Cancel' and 'Create' buttons. To the right of the dialog box, there is a table of installed packages with columns for Name, Description, and Version. The table lists several packages including aext-shared, aext_assistant, aext_assistant_ser..., aext_core, and Anaconda extensions shared library. The text '567 packages available' is visible at the bottom of the table. Red text annotations are present: 'Type environment name' pointing to the Name field and 'Select version' pointing to the Python version dropdown.

1

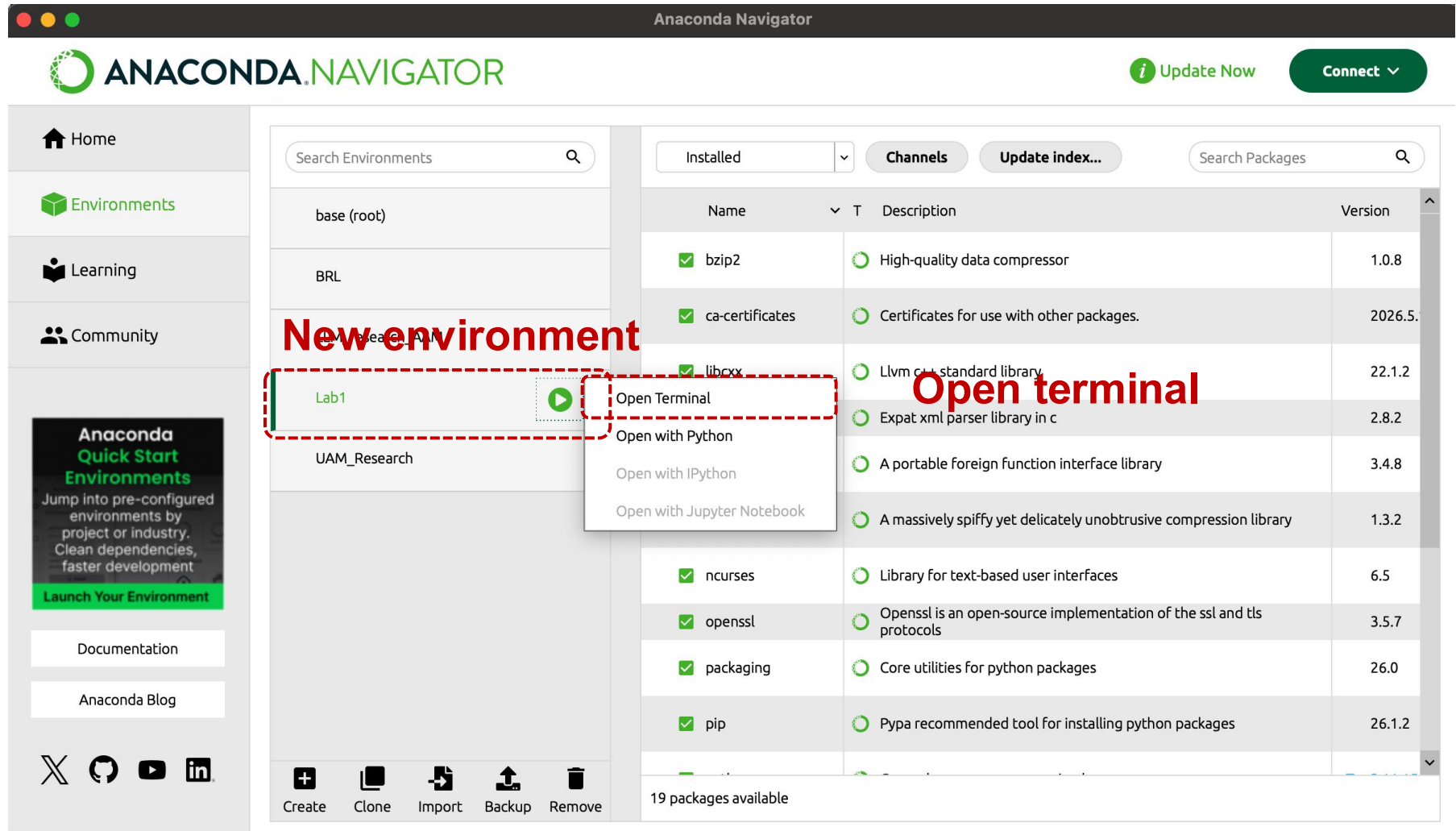
2

Type environment name

Select version

TensorFlow & Keras Setup

□ Step 4. Open Terminal and Activate the Environment



The screenshot shows the Anaconda Navigator interface. The left sidebar contains navigation links: Home, Environments, Learning, and Community. The main panel displays the 'Environments' tab, showing a list of environments: 'base (root)', 'BRL', 'Lab1', and 'UAM_Research'. The 'Lab1' environment is highlighted with a red dashed box. A red arrow points to the 'Open Terminal' button in the context menu. The text 'New environment' is written in red above the 'Lab1' environment, and 'Open terminal' is written in red next to the 'Open Terminal' button.

New environment

Open terminal

Search Environments

base (root)

BRL

Lab1

UAM_Research

Open Terminal

Open with Python

Open with IPython

Open with Jupyter Notebook

Installed

Channels

Update index...

Search Packages

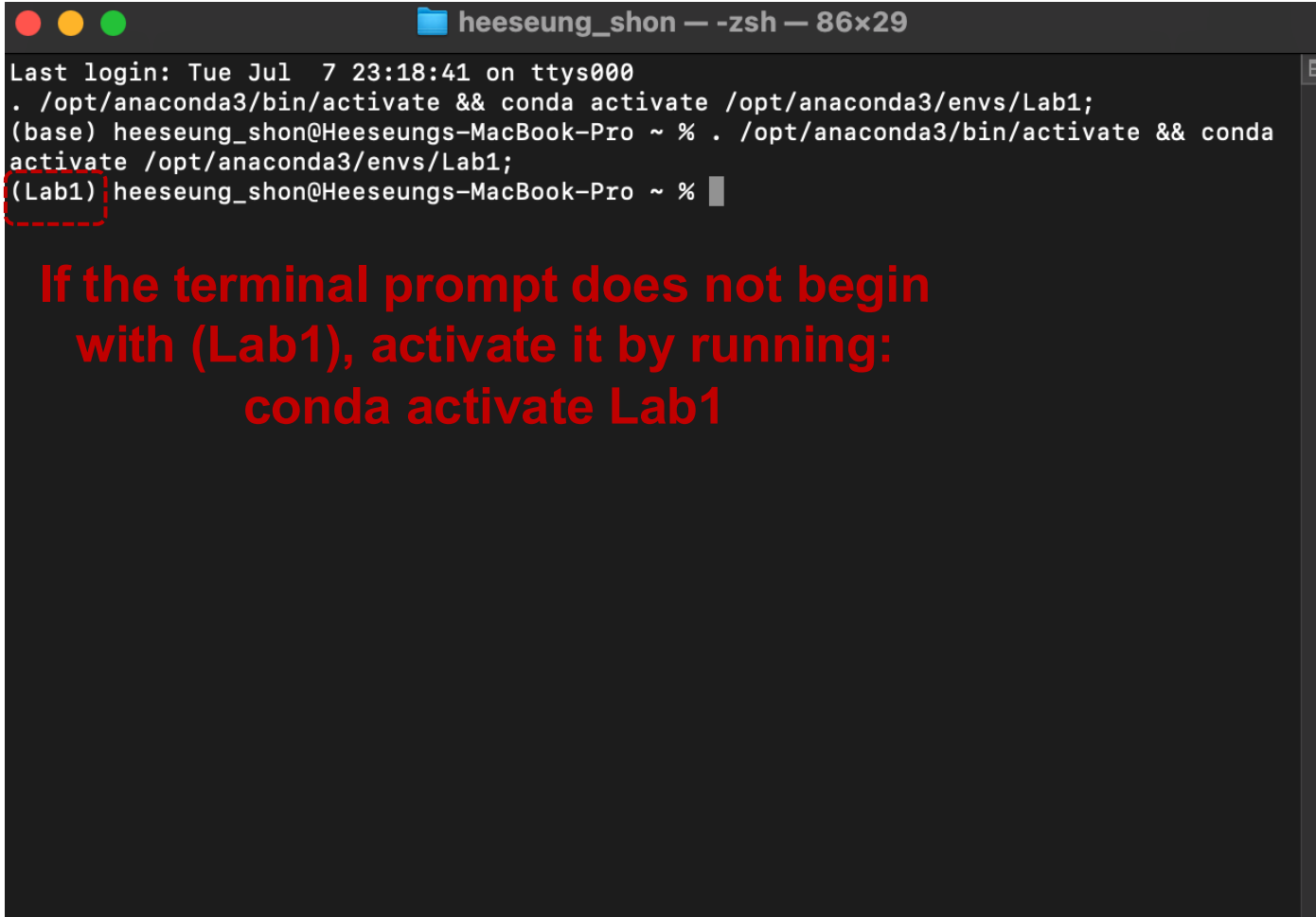
Name	T	Description	Version
✓ bzip2	○	High-quality data compressor	1.0.8
✓ ca-certificates	○	Certificates for use with other packages.	2026.5.
✓ libxx	○	Llvm c++ standard library	22.1.2
○	○	Expat xml parser library in c	2.8.2
○	○	A portable foreign function interface library	3.4.8
○	○	A massively spiffy yet delicately unobtrusive compression library	1.3.2
✓ ncurses	○	Library for text-based user interfaces	6.5
✓ openssl	○	Openssl is an open-source implementation of the ssl and tls protocols	3.5.7
✓ packaging	○	Core utilities for python packages	26.0
✓ pip	○	Pypa recommended tool for installing python packages	26.1.2

19 packages available

TensorFlow & Keras Setup

❑ Step 4. Open Terminal and Activate the Environment

Terminal

A terminal window titled "heeseung_shon — -zsh — 86x29" with standard macOS window controls (red, yellow, green buttons). The terminal shows the following commands and output:

```
Last login: Tue Jul 7 23:18:41 on ttys000
. /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(base) heeseung_shon@Heeseungs-MacBook-Pro ~ % . /opt/anaconda3/bin/activate && conda
activate /opt/anaconda3/envs/Lab1;
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ %
```

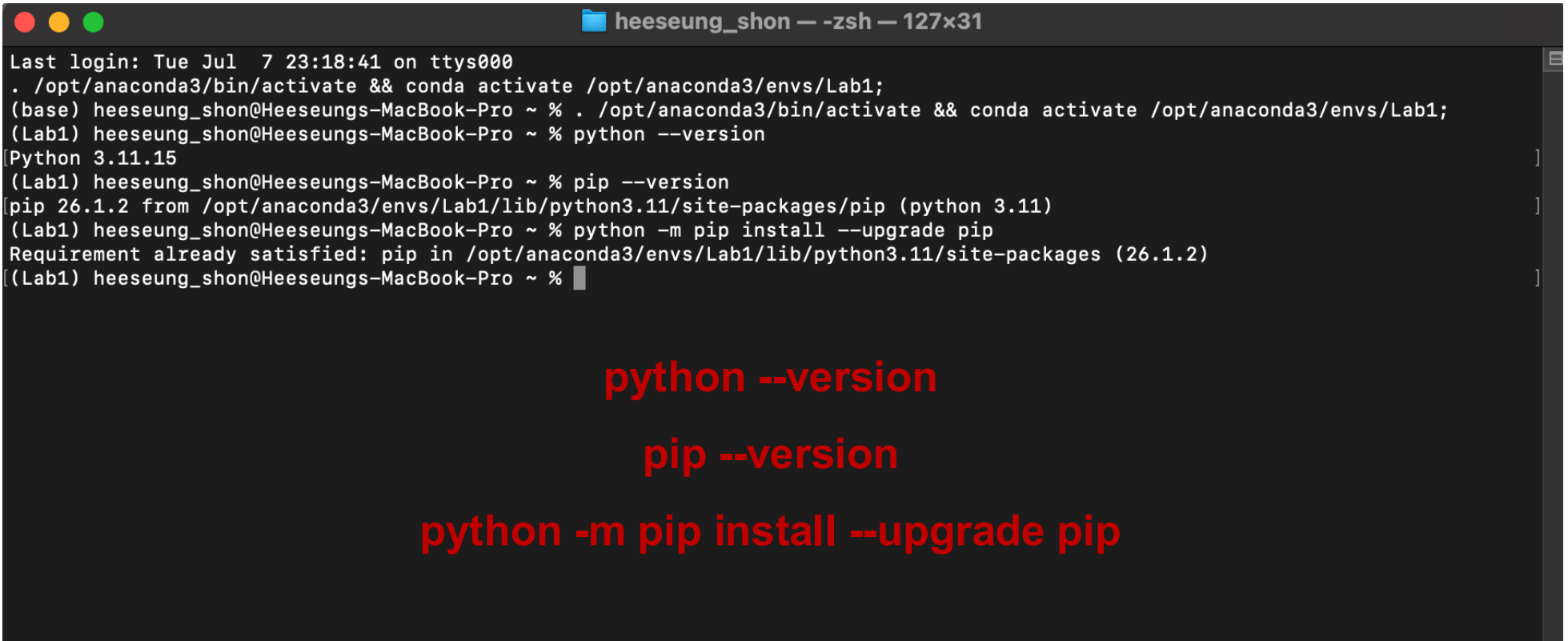
The prompt "(Lab1)" is highlighted with a red dashed box. Below the terminal output, there is a red text instruction: "If the terminal prompt does not begin with (Lab1), activate it by running: conda activate Lab1".

```
If the terminal prompt does not begin
with (Lab1), activate it by running:
conda activate Lab1
```

TensorFlow & Keras Setup

❑ Step 5. Check Python and pip

- ❖ Confirm that Python and pip are available in Lab1 before installing TensorFlow.



```
heeseung_shon — zsh — 127x31
Last login: Tue Jul 7 23:18:41 on ttys000
. /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(base) heeseung_shon@Heeseungs-MacBook-Pro ~ % . /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % python --version
Python 3.11.15
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % pip --version
pip 26.1.2 from /opt/anaconda3/envs/Lab1/lib/python3.11/site-packages/pip (python 3.11)
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % python -m pip install --upgrade pip
Requirement already satisfied: pip in /opt/anaconda3/envs/Lab1/lib/python3.11/site-packages (26.1.2)
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ %
```

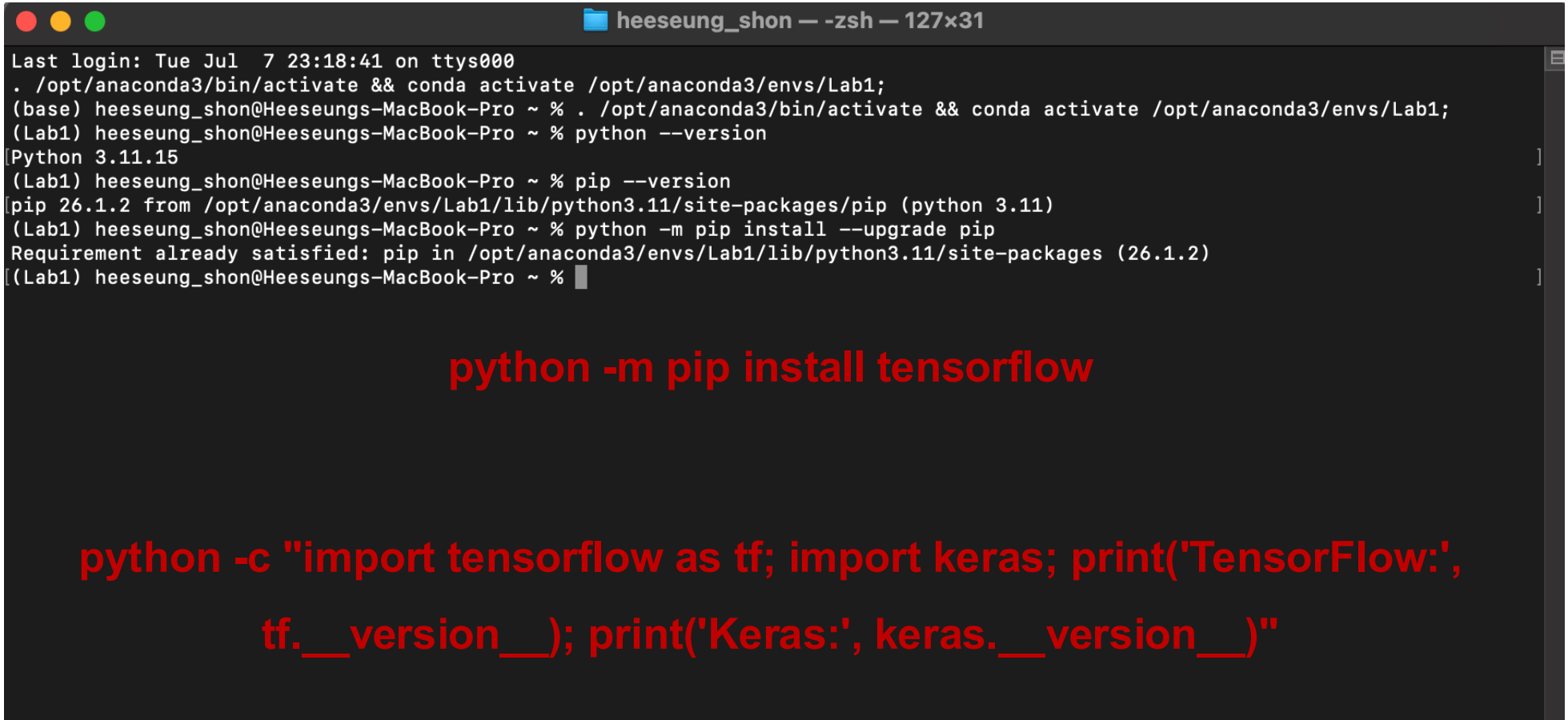
python --version

pip --version

python -m pip install --upgrade pip

TensorFlow & Keras Setup

❑ Step 6. Install and Verify TensorFlow & Keras



```
heeseung_shon — -zsh — 127x31
Last login: Tue Jul 7 23:18:41 on ttys000
. /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(base) heeseung_shon@Heeseungs-MacBook-Pro ~ % . /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % python --version
Python 3.11.15
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % pip --version
pip 26.1.2 from /opt/anaconda3/envs/Lab1/lib/python3.11/site-packages/pip (python 3.11)
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % python -m pip install --upgrade pip
Requirement already satisfied: pip in /opt/anaconda3/envs/Lab1/lib/python3.11/site-packages (26.1.2)
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ %
```

python -m pip install tensorflow

python -c "import tensorflow as tf; import keras; print('TensorFlow:', tf.__version__); print('Keras:', keras.__version__)"

TensorFlow & Keras Setup

❑ Step 7. Alternative: Install Packages in Navigator

The screenshot shows the Anaconda Navigator interface. On the left sidebar, the 'Environments' tab is selected, and 'Lab1' is highlighted with a red dashed box and the number 1. The main panel shows the 'Channels' view with a search bar containing 'tensorflow' (highlighted with a red dashed box and the number 3). Below the search bar, a list of packages is displayed. The 'tensorflow' package is selected with a green checkmark (highlighted with a red dashed box and the number 4). At the bottom right, the 'Apply' button is highlighted with a red dashed box and the number 5. The status bar at the bottom indicates '15 packages available matching "tensorflow" 2 packages selected'.

Name	Description	Version
tensorboard	Tensorflow's visualization toolkit	2.20.0
☒ tensorflow	Tensorflow is an end-to-end open source platform for machine learning.	2.21.0
tensorflow-base	Tensorflow is an end-to-end open source platform for machine learning.	2.21.0
tensorflow-cpu	Tensorflow is an end-to-end open source platform for machine learning.	2.21.0
☒ tensorflow-datasets	Tensorflow/datasets is a library of datasets ready to use with tensorflow.	4.9.10
tensorflow-estimator	Tensorflow is an end-to-end open source platform for machine learning.	2.17.0
tensorflow-hub	A library for transfer learning by reusing parts of tensorflow models.	0.16.1
tensorflow-metadata	Library and standards for schema and statistics.	1.17.3
tensorflow-probability	Tensorflow probability is a library for probabilistic reasoning and statistical analysis in tensorflow	0.24.0
tensorflow-text	Tensorflow is a machine learning library, base package contains only tensorflow.	2.20.1

TensorFlow & Keras Setup

❑ Step 8. Install Common Packages

- ❖ NumPy and SciPy for numerical computing.
- ❖ pandas for data handling.
- ❖ matplotlib for plotting.
- ❖ scikit-learn for machine learning basics.

```
heeseung_shon — -zsh — 127x31
Last login: Tue Jul  7 23:18:41 on ttys000
. /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(base) heeseung_shon@Heeseungs-MacBook-Pro ~ % . /opt/anaconda3/bin/activate && conda activate /opt/anaconda3/envs/Lab1;
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % python --version
Python 3.11.15
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % pip --version
pip 26.1.2 from /opt/anaconda3/envs/Lab1/lib/python3.11/site-packages/pip (python 3.11)
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ % python -m pip install --upgrade pip
Requirement already satisfied: pip in /opt/anaconda3/envs/Lab1/lib/python3.11/site-packages (26.1.2)
(Lab1) heeseung_shon@Heeseungs-MacBook-Pro ~ %
```

python -m pip install numpy scipy pandas matplotlib scikit-learn

TensorFlow & Keras Setup

❑ Step 9. Install Jupyter Notebook

The screenshot displays the Anaconda Navigator application window. On the left sidebar, the 'Home' button is highlighted with a red dashed box and the number '1'. The main panel shows a grid of application tiles. The 'Lab1' channel is selected in the top dropdown, which is also highlighted with a red dashed box and the number '2'. In the center of the grid, the 'Jupyter Notebook' tile is highlighted with a red dashed box and the number '3'. This tile shows the Jupyter logo, version 7.5.6, and an 'Install' button. Other visible tiles include 'Glueviz' (version 1.4.0) and 'Orange 3' (version 3.40.0), each with a 'Launch' button at the top and an 'Install' button at the bottom. The bottom row of the grid shows partial views of 'IP[y]:' and another application tile. The top right of the window features an 'Update Now' button and a 'Connect' button.

TensorFlow & Keras Setup

❑ Step 10. Launch Jupyter Notebook

